Bulk Issuance Context Management

Self-enforcing Data Loyalty

https://kericonf.com

Samuel M. Smith PhD
sam@kerisuite.org
https://keri.foundation

Duty of Data Loyalty

https://scholarship.law.bu.edu/faculty_scholarship/3143/

The idea behind loyalty is simple: organizations should not process data or design technologies that conflict with the best interests of trusting parties. ...

Our model takes advantage of loyalty's strengths—it is well-established in our law, it is flexible, and it can accommodate conflicting values. ...

But loyalty has one great virtue: it places the focus for information age problems on the relationships that define our social lives rather than on the data which is the byproduct of those relationships. ...

Data loyalty has three key features—it is a

(1) relational duty

(2) that prohibits self-dealing

(3) at the expense of a trusting party.

Enforcement via Relationship Loyalty

https://scholarship.law.bu.edu/faculty_scholarship/3143/

Finally, relationships open the possibility of more robust enforcement rules because they are voluntarily entered into and hold a unique place in the law as a result.

The concept of contractual privity could also be used to extend relational duties beyond the initial trusting party and entrustee.

Under a "chain-link" approach to relational privacy rules, lawmakers could directly—or using mandated terms in data-sharing contracts—link the disclosure of personal information to obligations of loyalty to protect information as it is disclosed downstream. To create the chain of protection, contracts would be used to link each new recipient of information to a previous recipient who wished to disclose the information.

At the same time, relational duties raise even fewer free expression issues than other forms of data regulation because they regulate relationships rather than information flows. In relationships, parties assume these duties by soliciting trust and voluntarily entering into these relationships.

Contract

A valid contract needs

1) an agreement,
2) an exchange of consideration, and
3) an intention to create legal relations.

ACDC Confidentiality/Privacy Protections

Least Disclosure:

The system should disclose only the minimum amount of information about a given party needed to facilitate a transaction, and no more

Graduated Disclosure: progressive least disclosures that each enable further disclosure

Contractually Protected Disclosure:

Uses graduated disclosure to minimize leakage prior to legally enforceable protection

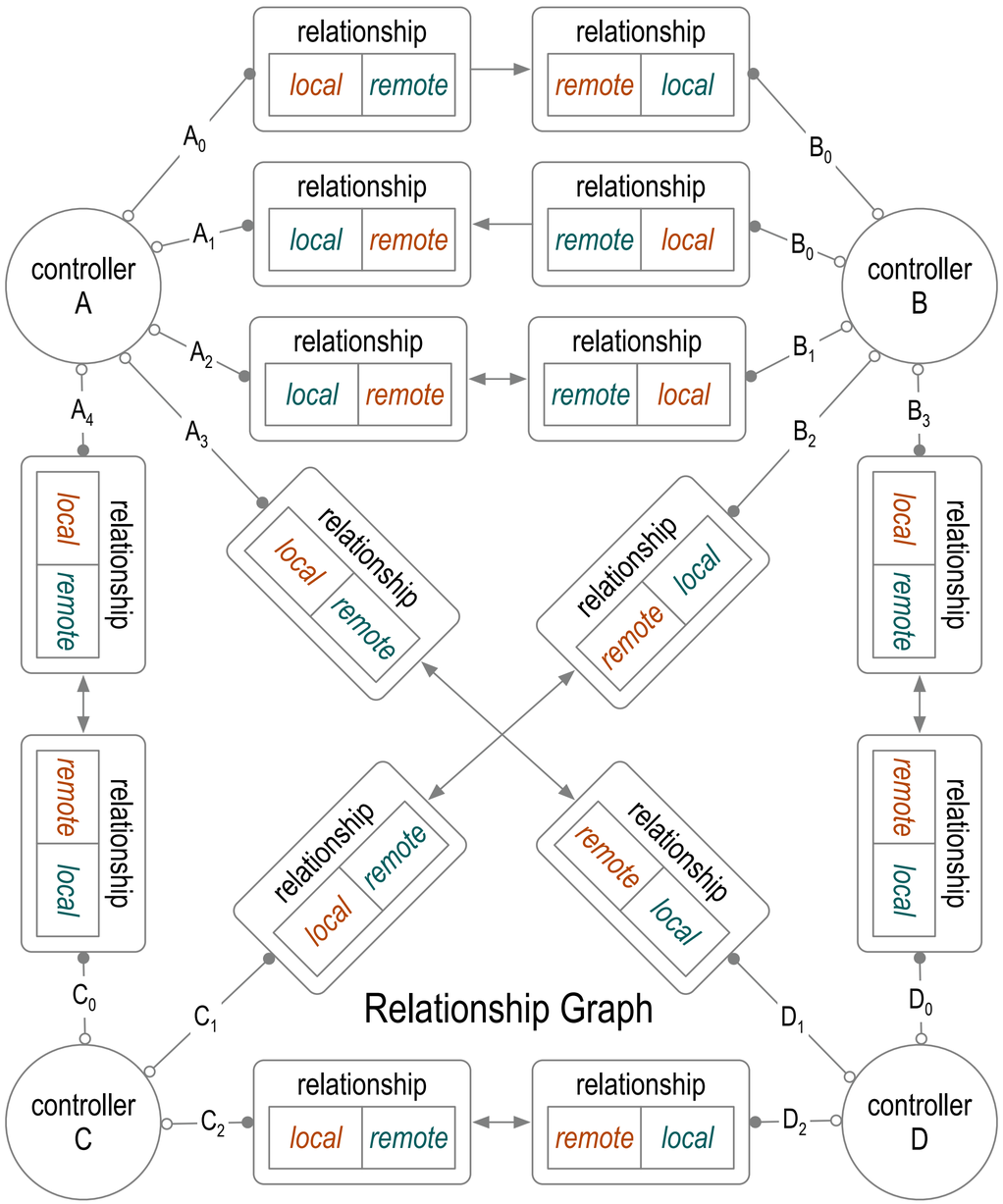
IPEX Protocol (issuance and presentation exchange)

Blindable ACDC State Registries

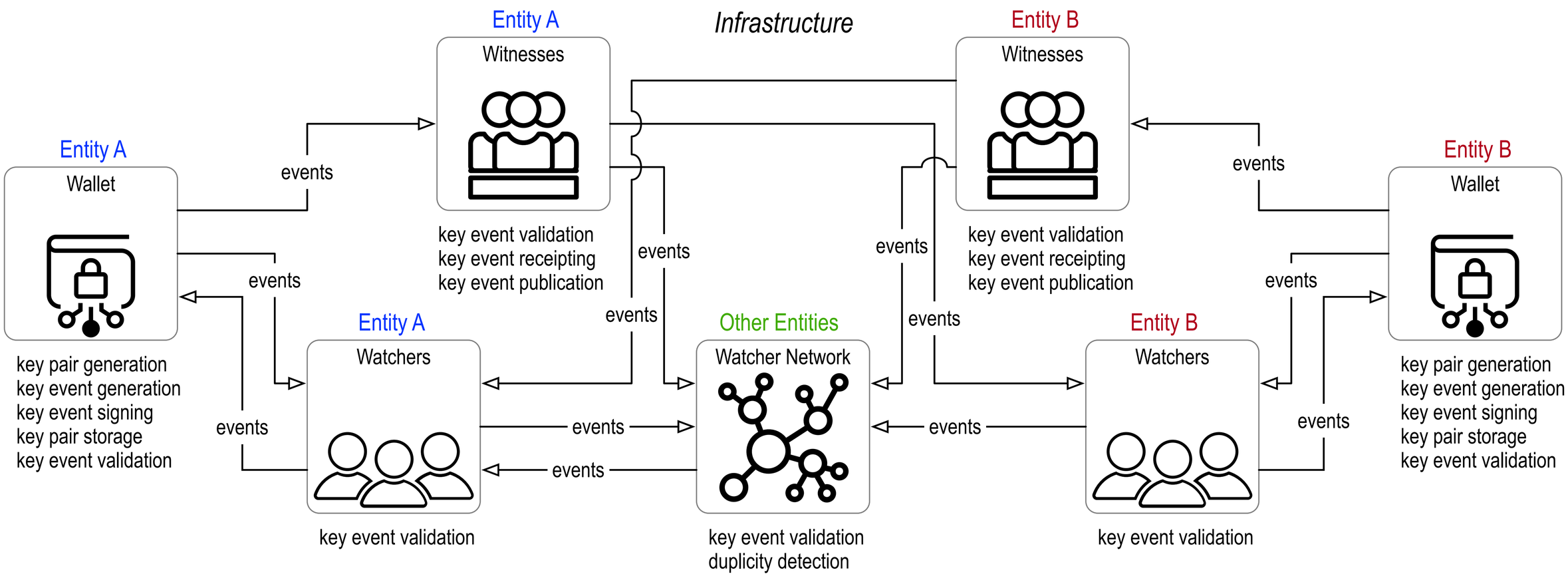
Bindable Blindable Delegated ACDC State Registries:

perpetually verifiable, dynamically revocable, delegable, transferable

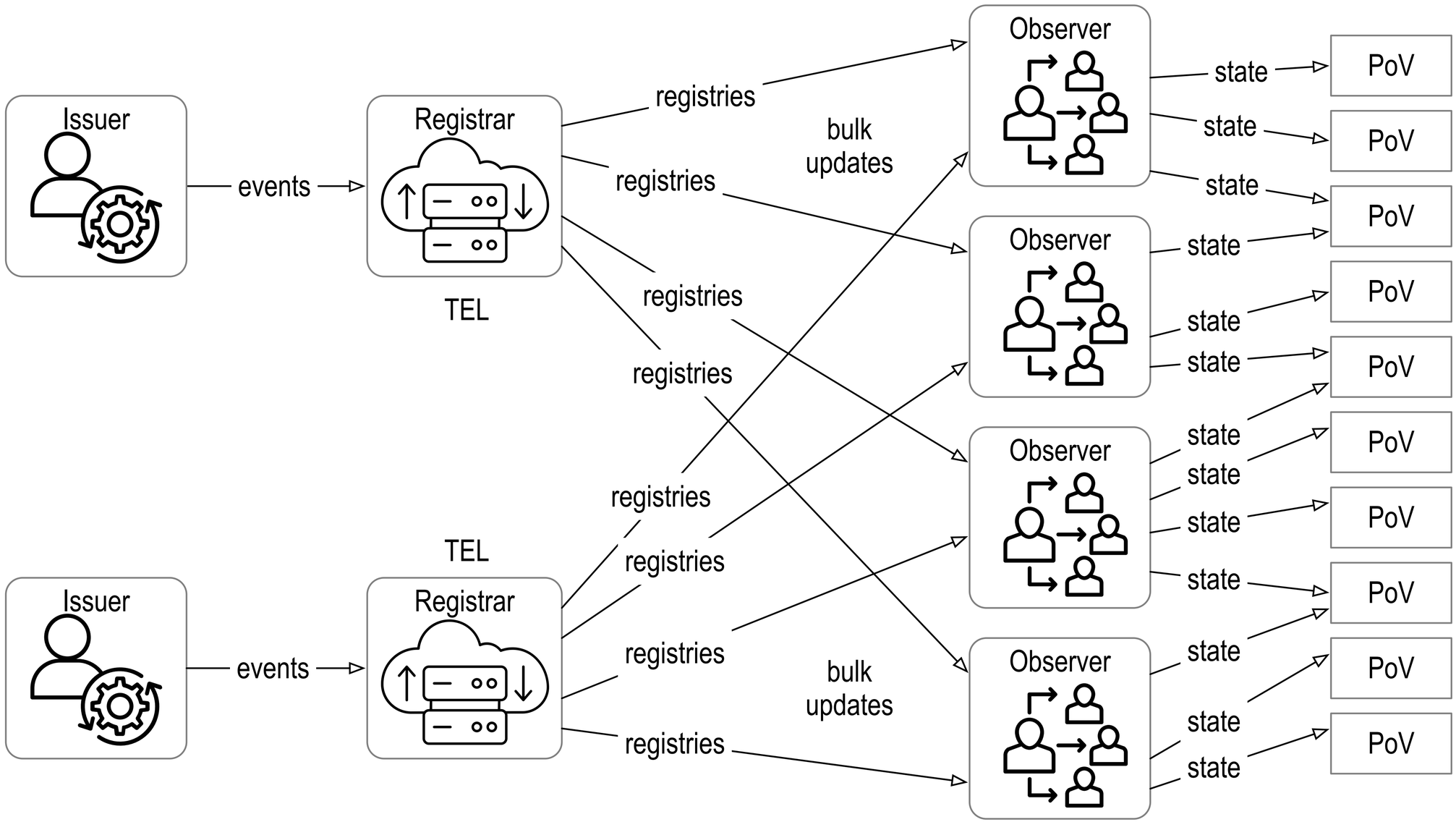
Relationship Graph (global)



Keri Infrastructure



Registrar and Observer



Issuance Registrar/Registries

ACDC

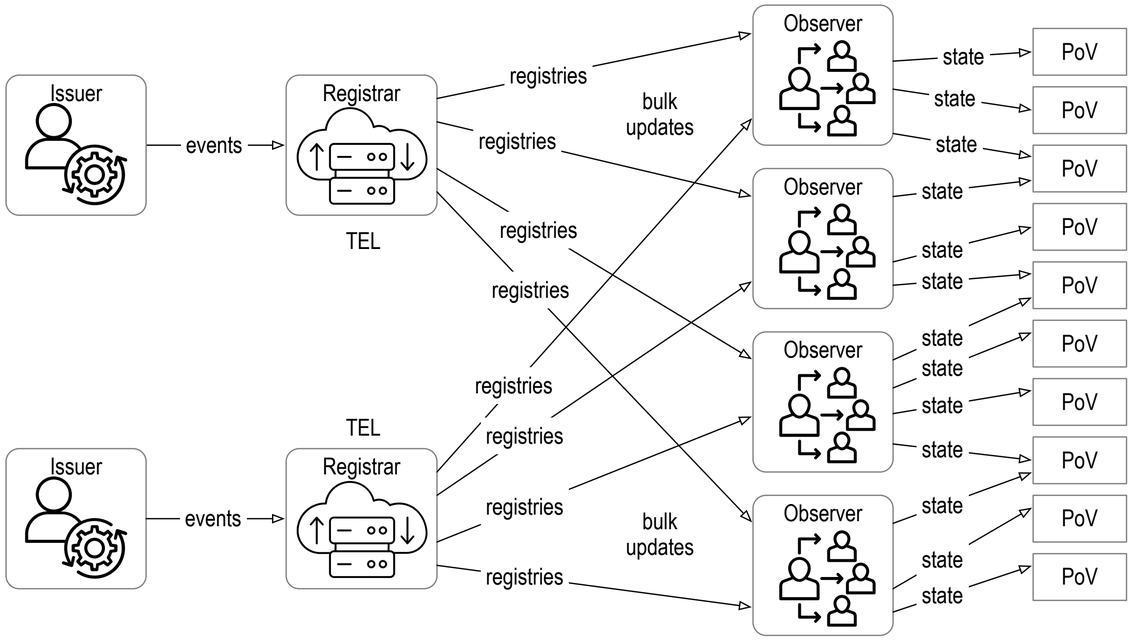
v	t	d	u	i	rd	s	a	e	r
Version	Type	SAID	UUID	Issuer AID	Registry	Schema	Attribute	Edge	Rule

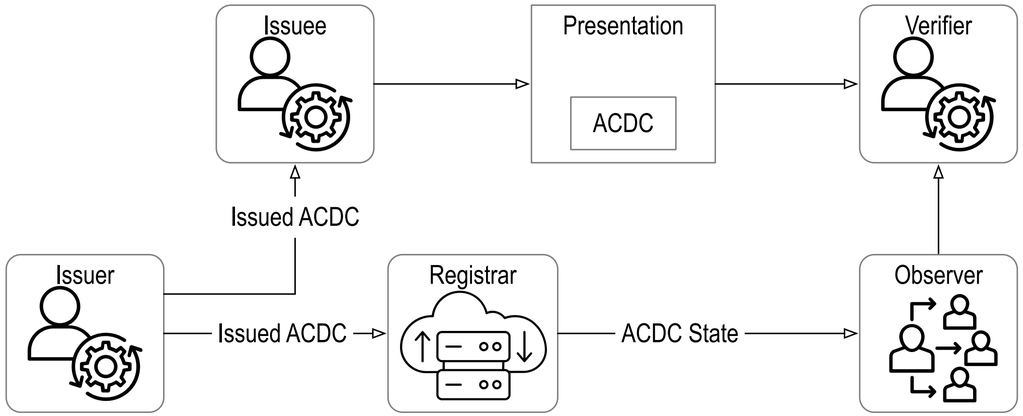
Registry Inception Event (rip)

v	t	d	u	i	n	dt
Version	Type	SAID	UUID	Issuer AID	SN	DateTime

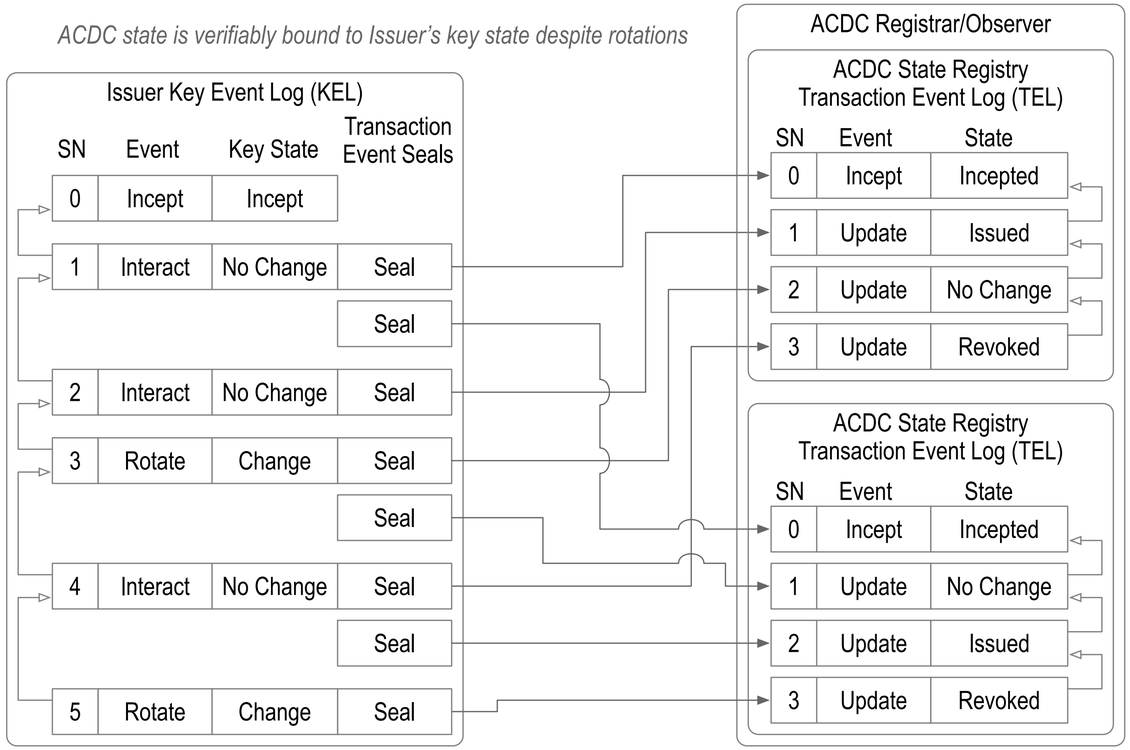
Registry Blinded Update Event (bup)

v	t	d	rd	n	p	dt	b
Version	Type	SAID	Registry	SN	Prior	DateTime	BLID

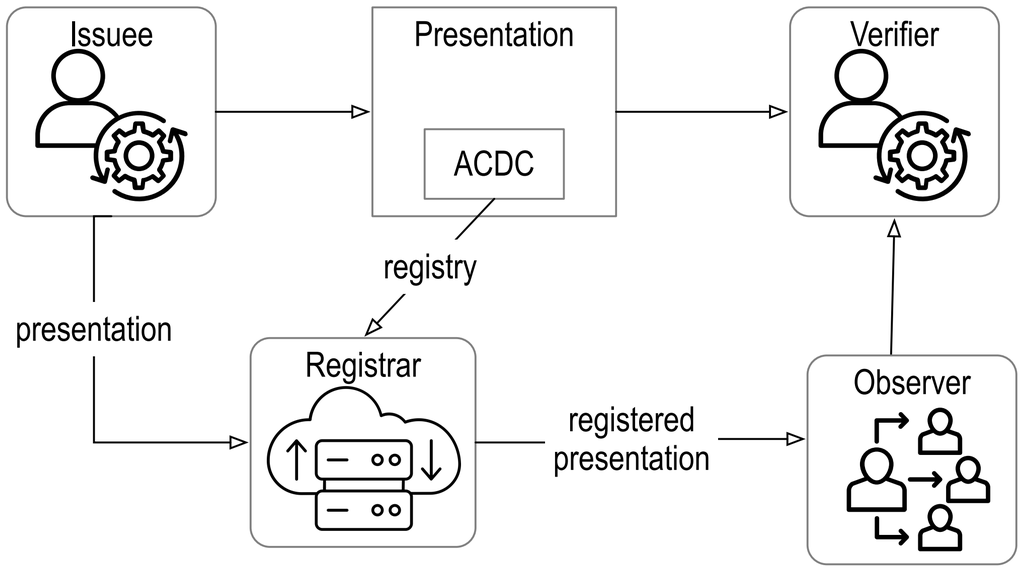




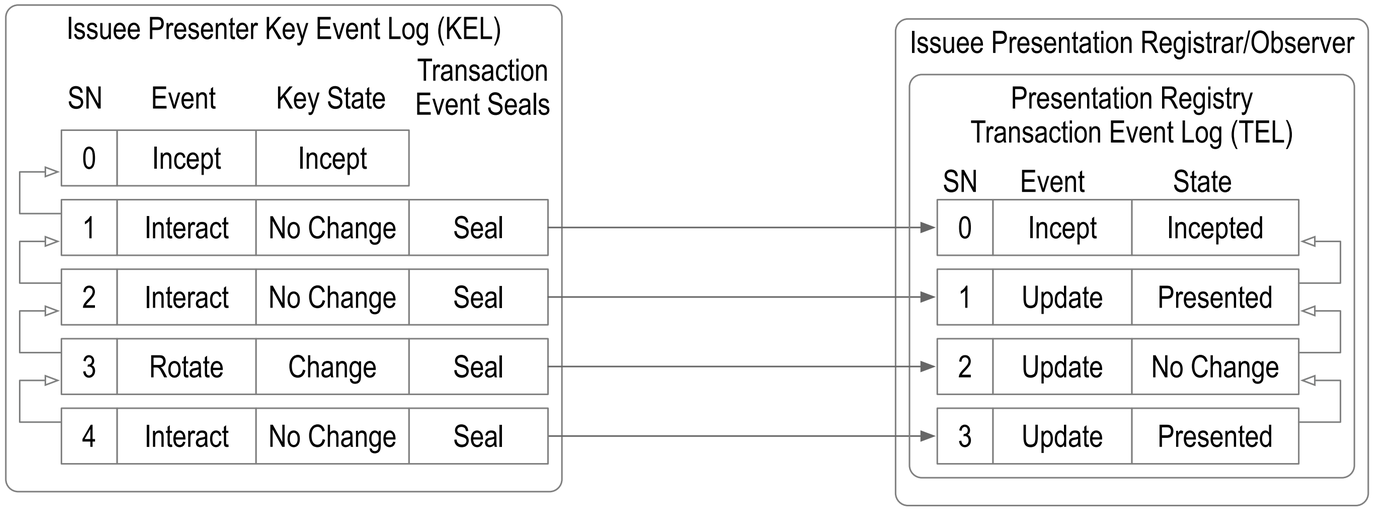
ACDC state is verifiably bound to Issuer's key state despite rotations



Usage Presentation Registry



ACDC Issuee Presentation state is verifiably bound to Issuee's key state despite rotations



ACDC Bulk Issuance with full cryptographic unlinkability validator to validator

The UUIDs in each member of a bulk-issued set are derived using an HDKey algorithm that includes a shared secret salt, the member index, and the UUID index.

Each member of a bulk-issued set has a unique AID and unique keys derived using an HDKey algorithm that includes a non-shared private salt, a shared secret salt, and the member index.

When using *independent registry bulk-issued ACDCs*, the total number of registries for a given set of bulk-issued ACDCs typically equals the number of members in that set.

Alternatively, a given Issuer could use one blinded state tel for multiple different ACDCs issued to the same issuee, thereby providing herd privacy of usage and state across a set of issuances.

ACDC Example

Cal Registry Inception


```
{
  "v": “ACDCCAACAAJSONAADa.",
  "t": "rip",
  "d": "EPtolmh_NE2vC02oFc7FOiWkPcEiKUPWm5uu_Gv1JZDw",
  "u": "0ABhY2Rjc3BlY3dvcmtyYXcy",
  "i": "ECsGDKWAYtHBCkiDrzajkxs3Iw2g-dls3bLUsRP4yVdT",
  "n": "0",
  "dt": "2025-07-04T17:52:00.000000+00:00"
}
```


Cal Issued Most Compact ACDC


```
{
  "v": "ACDCCAACAAJSONAAF3.",
  "t": "acm",
  "d": "EIF7egPvC8ITbGRdM9G0kd6aPELDg-azMkAqT-7cMuAi",
  "u": "0ABhY2Rjc3BlY3dvcmtyYXdh",
  "i": "ECsGDKWAYtHBCkiDrzajkxs3Iw2g-dls3bLUsRP4yVdT",
  "rd": "EPtolmh_NE2vC02oFc7FOiWkPcEiKUPWm5uu_Gv1JZDw",
  "s": "EK_iGlfdc7Q-qIGL-kqbDSD2z4fesT4dAQLEHGgH4lLG",
  "a": "EK799owRYyk8UPFWUmfsm5AJfJmU7jZGtZXJFbg2I0KL",
  "r": "EMZf9m0XYwqo4L8tnIDMZuX7YCZnMswS7Ta9j0CuYfjU"
}
```


Cal Issued ACDC


```
{
  "v": "ACDCCAACAAJSONAAKX.",
  "t": "acm",
  "d": "EIF7egPvC8ITbGRdM9G0kd6aPELDg-azMkAqT-7cMuAi",
  "u": "0ABhY2Rjc3BlY3dvcmtyYXdh",
  "i": "ECsGDKWAYtHBCkiDrzajkxs3Iw2g-dls3bLUsRP4yVdT",
  "rd": "EPtolmh_NE2vC02oFc7FOiWkPcEiKUPWm5uu_Gv1JZDw",
  "s": "EK_iGlfdc7Q-qIGL-kqbDSD2z4fesT4dAQLEHGgH4lLG",
  "a":
  {
    "d": "EK799owRYyk8UPFWUmfsm5AJfJmU7jZGtZXJFbg2I0KL",
    "u": "0ABhY2Rjc3BlY3dvcmtyYXc3",
    "i": "ECmiMVHTfZIjhA_rovnfx73T3G_FJzIQtzDn1meBVLAz",
    "name": "Sunspot College",
    "level": "gold"
  },
  "r":
  {
    "d": "EMZf9m0XYwqo4L8tnIDMZuX7YCZnMswS7Ta9j0CuYfjU",
    "l": "Issuer provides this ACDC on an AS IS basis. This ACDC in
whole or in part MUST NOT be shared with any other entity besides the
intended recipient."
  }
}
```

ACDC Selective Disclosure Example

Aggregate Section


```
"A":
[
  "EN5d44fTNM0M4kmMMVrsH0HwMLRLyb6SoJEV0ogkLdXx",
  {
      "d": "EI2lwi1ZKrs-bDwgEreOhEh-W2O5xrOm5T-QCyMuX5V4",
      "u": "0ABhY2Rjc3BlY3dvcmtyYXcw",
      "i": "ECWJZFBtllh99fESUOrBvT3EtBujWtDKCmyzDAXWhYmf"
  },
  {
      "d": "EC-vU19URXX8ztfWdp_j2HHr1lJsqtGa1YHtZrg6-GMR",
      "u": "0ABhY2Rjc3BlY3dvcmtyYXcx",
      "score": 96
  },
  {
      "d": "EKYLUIpDXNT0ujSdoNOT5pLp0okOKW3mAbg-M7K5OO_C",
      "u": "0ABhY2Rjc3BlY3dvcmtyYXcy",
      "name": "Zoe Doe"
  }
]
```


Blinded Aggregate Element List


```
"A":
[
  "EN5d44fTNM0M4kmMMVrsH0HwMLRLyb6SoJEV0ogkLdXx",
  "EI2lwi1ZKrs-bDwgEreOhEh-W2O5xrOm5T-QCyMuX5V4",
  "EC-vU19URXX8ztfWdp_j2HHr1lJsqtGa1YHtZrg6-GMR",
  "EKYLUIpDXNT0ujSdoNOT5pLp0okOKW3mAbg-M7K5OO_C"
]
```


AGID = H(C(a_i for all i in \{0, ..., N\})), where

H is the digest (hash) Operator and

C is the ordered serialization operator.

This aggregate AGID makes a blinded cryptographic commitment to all SAIDS in the list [a_1, …, a_N].

"A": “EN5d44fTNM0M4kmMMVrsH0HwMLRLyb6SoJEV0ogkLdXx"

ACDC Selective Disclosure Example …

Selectively Disclosed Aggregate Section


```
"A":
[
  "EN5d44fTNM0M4kmMMVrsH0HwMLRLyb6SoJEV0ogkLdXx",
  {
    "d": "EI2lwi1ZKrs-bDwgEreOhEh-W2O5xrOm5T-QCyMuX5V4",
    "u": "0ABhY2Rjc3BlY3dvcmtyYXcw",
    "i": "ECWJZFBtllh99fESUOrBvT3EtBujWtDKCmyzDAXWhYmf"
  },
  {
    "d": "EC-vU19URXX8ztfWdp_j2HHr1lJsqtGa1YHtZrg6-GMR",
    "u": "0ABhY2Rjc3BlY3dvcmtyYXcx",
    "score": 96
  },
  "EKYLUIpDXNT0ujSdoNOT5pLp0okOKW3mAbg-M7K5OO_C"
]
```

Blinded State Registry

ACDC

v	t	d	u	i	rd	s	a	e	r
Version	Type	SAID	UUID	Issuer AID	Registry	Schema	Attribute	Edge	Rule

Registry Inception Event (rip)

v	t	d	u	i	n	dt
Version	Type	SAID	UUID	Issuer AID	SN	DateTime

Registry Blinded Update Event (bup)

v	t	d	rd	n	p	dt	b
Version	Type	SAID	Registry	SN	Prior	DateTime	BLID

Blinded Attribute Block (BAB)

d	u	td	ts
BLID	UUID	ACDC SAID	State

Blinded State Registry (TEL)

Issuer Key Event Log (KEL)

SN	Event	Key State	Transaction Event Seals
0	Incept	Incept	
1	Interact	No Change	Seal
2	Interact	No Change	Seal
3	Rotate	Change	Seal
4	Interact	No Change	Seal
5	Rotate	Change	Seal
6	Rotate	Change	Seal

ACDC Registrar/Observer

ACDC State Registry Transaction Event Log (TEL)

SN	Event	BLID
0	Incept	…
1	Update	…
2	Update	…
3	Update	…
4	Update	…
5	Update	…

Blinded Attribute Blocks (BABs)

BLID	UUID	empty	empty
BLID	UUID	ACDC	issued
BLID	UUID	ACDC	issued
BLID	UUID	ACDC	revoked
BLID	UUID	ACDC	revoked

Shared Blinded State Registry

ACDC

v	t	d	u	i	rd	s	a	e	r
Version	Type	SAID	UUID	Issuer AID	Registry	Schema	Attribute	Edge	Rule

Registry Inception Event (rip)

v	t	d	u	i	n	dt
Version	Type	SAID	UUID	Issuer AID	SN	DateTime

Registry Blinded Update Event (bup)

v	t	d	rd	n	p	dt	b
Version	Type	SAID	Registry	SN	Prior	DateTime	BLID

Blinded Attribute Block (BAB)

d	u	td	ts
BLID	UUID	ACDC SAID	State

Issuer Key Event Log (KEL)

SN	Event	Key State	Transaction Event Seals
0	Incept	Incept	
1	Interact	No Change	Seal
2	Interact	No Change	Seal
3	Rotate	Change	Seal
4	Interact	No Change	Seal
5	Rotate	Change	Seal
6	Rotate	Change	Seal

Blinded State Registry (TEL)

ACDC Registrar/Observer

ACDC State Registry Transaction Event Log (TEL)

SN	Event	BLID
0	Incept	…
1	Update	…
2	Update	…
3	Update	…
4	Update	…
5	Update	…

Blinded Attribute Blocks (BABs)

BLID	UUID	empty	empty
BLID	UUID	ACDC1	issued
BLID	UUID	ACDC2	issued
BLID	UUID	ACDC2	revoked
BLID	UUID	ACDC1	revoked

ACDC Blinded Registry Example

Registry Inception


```
{
 "v": "ACDCCAACAAJSONAADa.",
  "t": "rip",
  "d": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
  "u": "0ABhY2Rjc3BlY3dvcmtyYXcx",
  "i": "ECWJZFBtllh99fESUOrBvT3EtBujWtDKCmyzDAXWhYmf",
  "n": "0",
  "dt": "2025-07-04T17:51:00.000000+00:00"
}
```


Registry Placeholder Update


```
{
  "v": "ACDCCAACAAJSONAAEi.",
  "t": "bup",
  "d": "EPNwyvHp2XJsz9pSpXtHtcCmzw6bKSFc-nhGKTbso0Yg",
  "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
  "n": "1",
  "p": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
  "dt": "2025-08-01T18:06:10.988921+00:00",
  "b": "ECVr7QWEp_aqVQuz4yprRFXVxJ-9uWLx_d6oDinlHU6J"
}
```


Virtual Label	Value	Description
`d`	`ECVr7QWEp_aqVQuz4yprRFXVxJ-9uWLx_d6oDinlHU6J`	BLID (Blinding SAID)
`u`	`aG1lSjdJSNl7TiroPl67Uqzd5eFvzmr6bPlL7Lh4ukv8`	UUID salty nonce blinding factor, random or HD generated
`td`	`1AAP`	Transaction ACDC SAID field value, top-level `d`
`ts`	`1AAP`	Transaction state value string

Unblinded attribute block as attachment to disclosure of TEL Event

`-aAYECVr7QWEp_aqVQuz4yprRFXVxJ-9uWLx_d6oDinlHU6JaG1lSjdJSNl7TiroPl67Uqzd5eFvzmr6bPlL7Lh4ukv81AAP1AAP`

Placeholder updates provide herd privacy wrt correlation to date time of actual state changes

ACDC Blinded Registry Example …

Registry Issued Update


```
{
  "v": "ACDCCAACAAJSONAAEi.",
  "t": "bup",
  "d": "EBdytzDC4dnatn-6mrCWLSGuM62LM0BgS31YnAg5NTeW",
  "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
  "n": "2",
  "p": "EPNwyvHp2XJsz9pSpXtHtcCmzw6bKSFc-nhGKTbso0Yg",
  "dt": "2020-08-02T12:00:20.000000+00:00",
  "b": "EOtWw6X_aoOJlkzNaLj23IC6MXHl7ZSYSWVulFW_Hr_t"
}
```


Virtual Label	Value	Description
`d`	`EOtWw6X_aoOJlkzNaLj23IC6MXHl7ZSYSWVulFW_Hr_t`	Dummied BLID (Blinding SAID)
`u`	`aLfCdNAnc-0P2SiruarZSajXiUWu5iU2VfQahvpNCyzB`	UUID salty nonce blinding factor, random or HD generated
`td`	`EMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4M`	Transaction ACDC SAID field value, top-level `d`
`ts`	`0Missued`	Transaction state value string

ACDC Blinded Registry Example …

Registry Issued Update


```
{
    "v": "ACDCCAACAAJSONAAEi.",
    "t": "bup",
    "d": "EM8B1uDhWaJLfpIiEqgp-3EurGUcbfe7u2k5AarDl2XD",
    "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
    "n": "3",
    "p": "EBdytzDC4dnatn-6mrCWLSGuM62LM0BgS31YnAg5NTeW",
    "dt": "2020-08-03T12:00:20.000000+00:00",
    "b": "EPj3sZj8OOWTkTgAN5vzVYdANeoj3zxgEn5APb8fCRRN"
}
```


Virtual Label	Value	Description
d	EPj3sZj8OOWTkTgAN5vzVYdANeoj3zxgEn5APb8fCRRN	Dummied BLID (Blinding SAID)
u	aGx7b16vGHVPT56tX30kYOEzTwiVY4aabc4k9AawYyZG	UUID salty nonce blinding factor HD generated
td	EMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4M	Transaction ACDC SAID field value, top-level d
ts	Yrevoked	Transaction state value string

Bulk Issuance

All the seals are anchored using a single seal, which is the Merkle root of a Sparse Merkle tree that references the transaction events for all the independent Registry TELs.

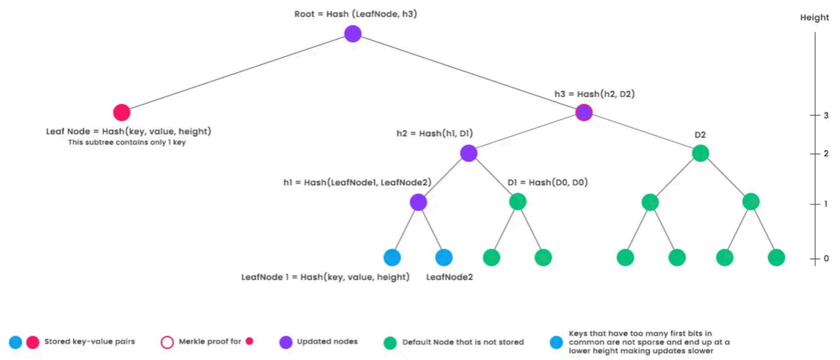
In a Sparse Merkle tree, the seals are only stored in leaf nodes of the tree, not in intermediate nodes

Sparse Merkle trees (SMTs) were originally developed for the Certificate Transparency (CT) network.

Since then, various optimized versions of SMT have been developed

Efficient inclusion proofs in an SMT enable a given Issuer to amalgamate all Registry transaction event seals for its bulk-issued ACDCs into a single SMT.

This provides so-called herd privacy to the updates.



Bulk Issuance Continued

To further strengthen herd privacy, when using blindable state Registries, the Issuer could randomly issue update events that do not change state, across all its registries, mixing updates over time and across each registry.

This would ensure that enough updates are captured by each batch anchoring seal to guarantee a pre-specified level of herd privacy.

A given seal in the KEL of the Issuer no longer provides a point of correlation to any other transaction event to any given registry.

A given inclusion proof for one event from one Registry does not reveal inclusion for events in other Registries or correlate a given bulk-issued ACDC to any other bulk-issued ACDC using a different Registry.

This amalgamation requires only one seal in the Issuer's KEL.

Using an optimized SMT algorithm enables an Issuer to provide efficient inclusion proofs for an amalgamated SMT of all transaction update events across all Registries for all bulk-issued ACDCs to all Issuees.

More Bulk Issuance

When using blindable state update events in these Registries, then the ACDC itself is not viewable by a 3rd party.

Because the SMT provides herd privacy with respect to which Registries with their transaction event updates belong to which bulk-issued sets of ACDCS, a pair of malicious 2nd parties cannot compare inclusion proofs to determine if two ACDCs belong to the same bulk-issued set.

Moreover, because each ACDC in a bulk-issued set gets its own Registry, the registry database does not provide a point of correlation between two different Registries that belong to the same bulk-issued set of ACDCs.

Therefore, a given disclosure of a bulk-issued ACDC does not directly disclose any points of correlation between malicious 2nd parties.

Three Party Information Sharing and Leakage Model

Information is Shared between two parties.

Third party observers may obtain information:

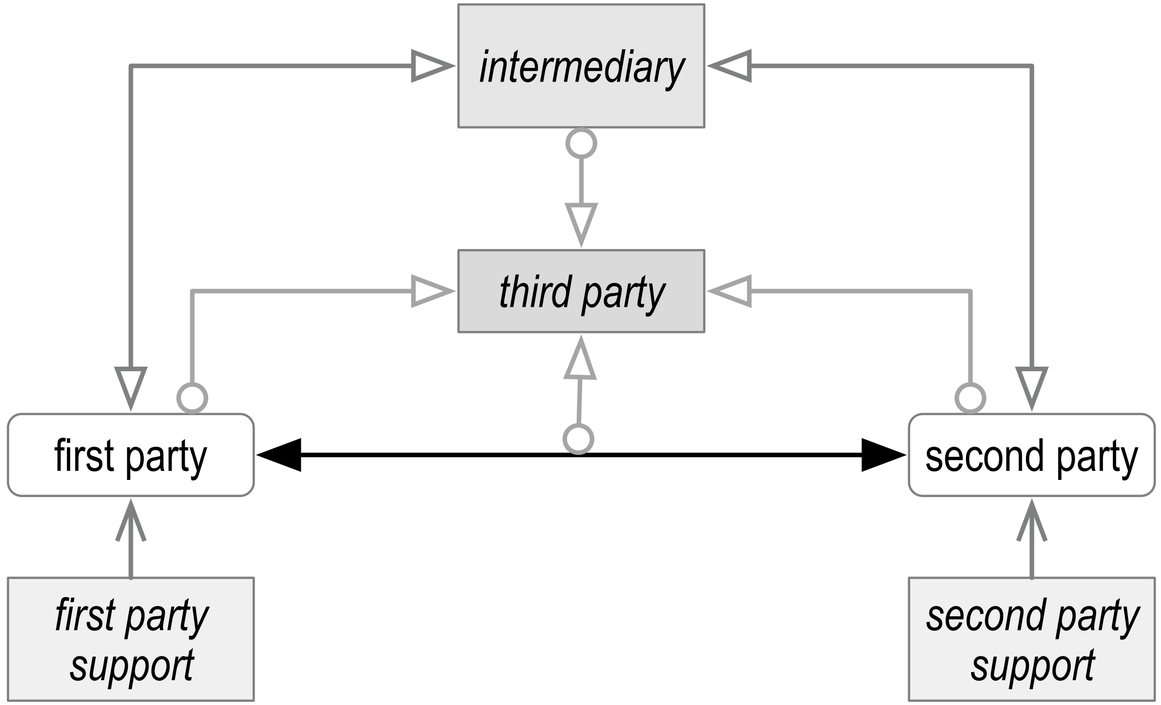
Indirectly as meta-data about the sharing interaction via leakage of the channel.

Directly from either of the first two parties and/or their supporting infrastructure including intermediaries.

Surveillance:

privacy is when third parties have no knowledge of ***who*** is sharing information

confidentiality is when third parties have no knowledge of ***what*** was shared.



Privacy requires accountability

Second parties to repeated interactions can correlate the first parties

Any interaction system that does not enable first parties to hold second parties accountable is fundamentally incompatible with privacy

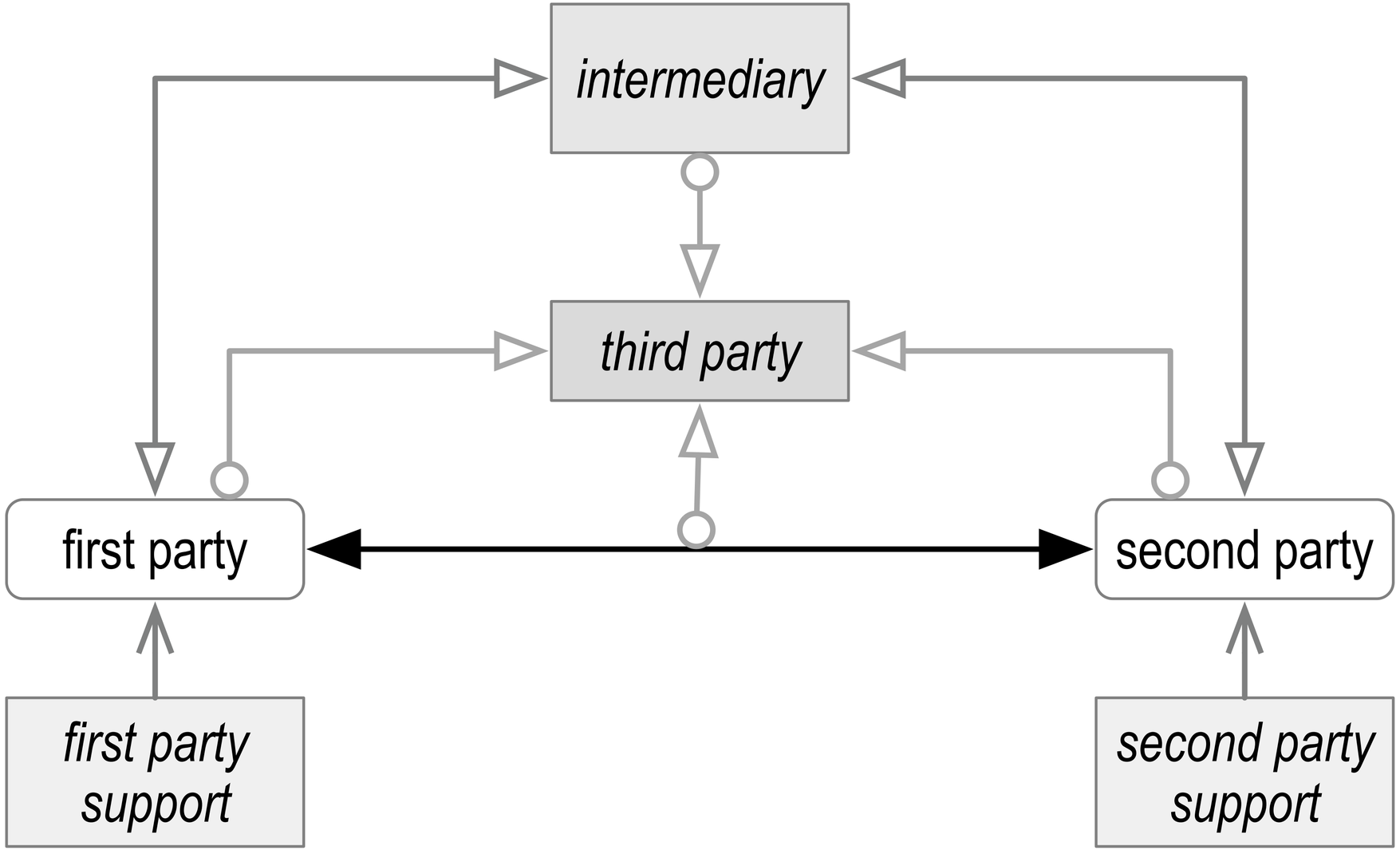
How to enforce accountability?

Control Confidential Contexts

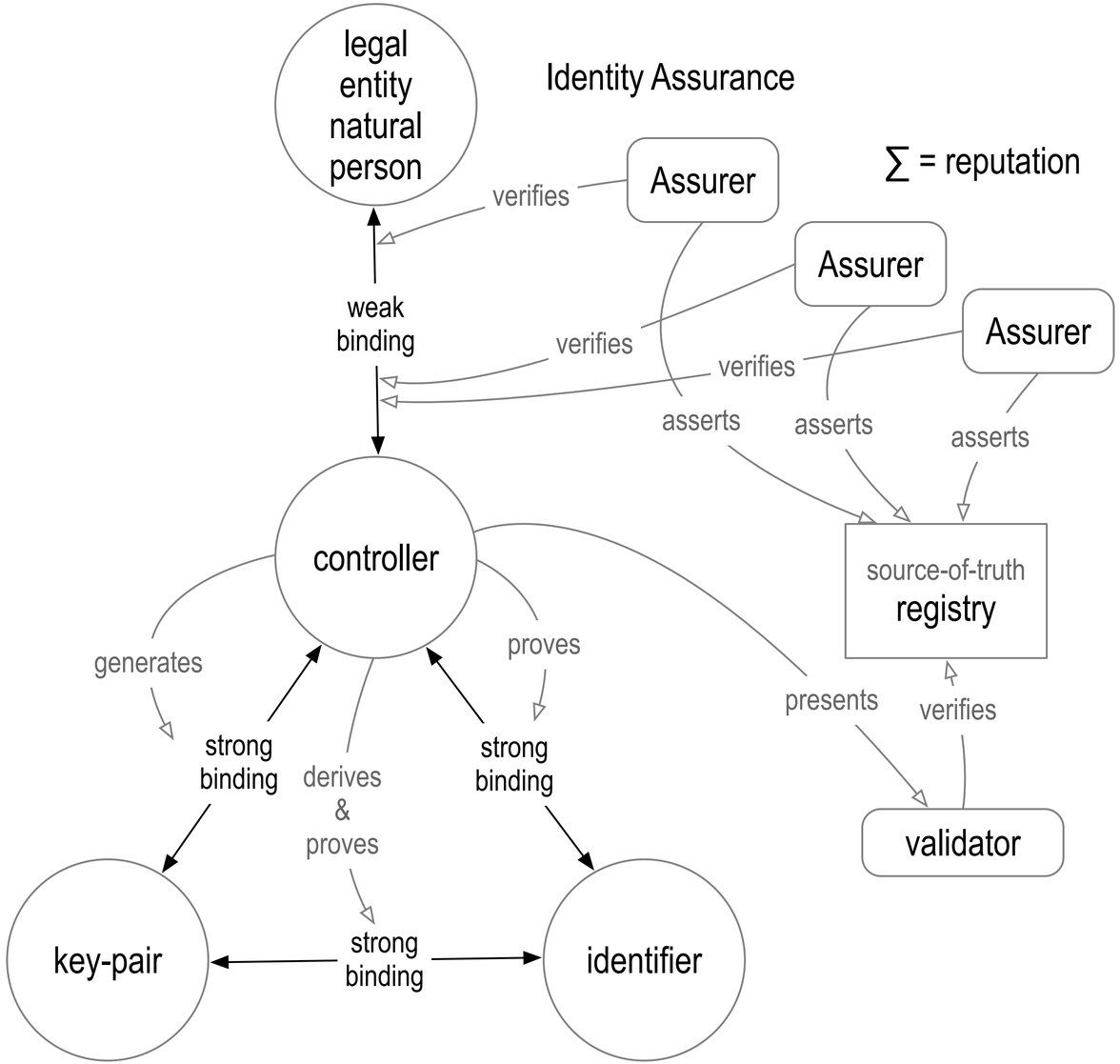
KERI Suite & SPAC/TSP

Create and control **safe** relationships where interactions are **protected** from **surveillance** by third parties and second parties are held **accountable** for any information shared **with** or **by** them

Three Party Information Sharing and Leakage Model



Identity Assurance and Reputation



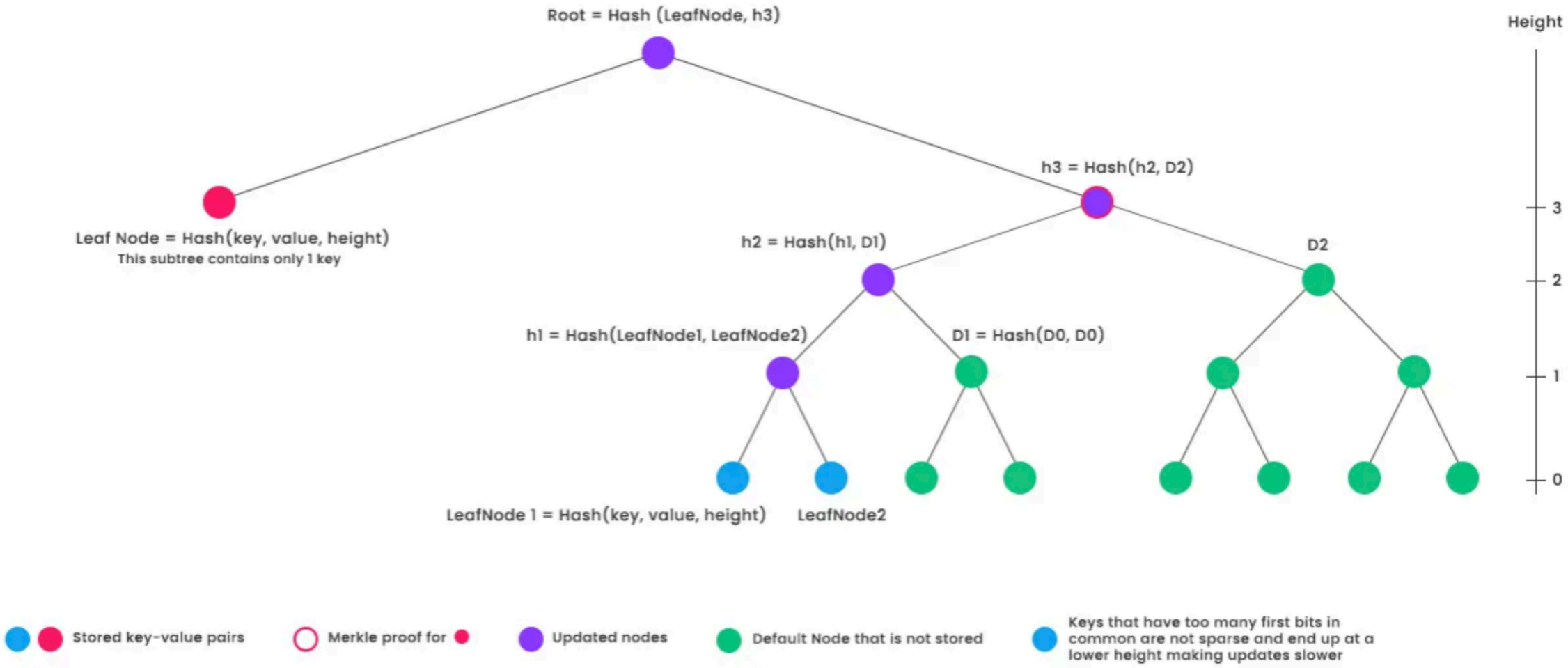
Relatively weak binding reinforced by multiple bindings = reputation

IAPC ACDC

An in-person session that binds identity assurance to cryptographic identity
NIST IAL3 Compliant
SEDI Account AID
MITM protected challenge response of proof of control over AID
Proofing Agent issues identity assurance proof-of-control ACDC to Account AID
IAPC include core attributes + biometric
Account AID is later presented to State to request bulk issuance of SEDI credentials
This enables dynamic sizing of SEDI bulk issuance sets

Sparse Merkle Tree

Optimized Compact Sparse Merkle Tree (all operations O(logN))



https://ouvrard-pierre-alain.medium.com/optimized-sparse-merkle-tree-fb481875d56e

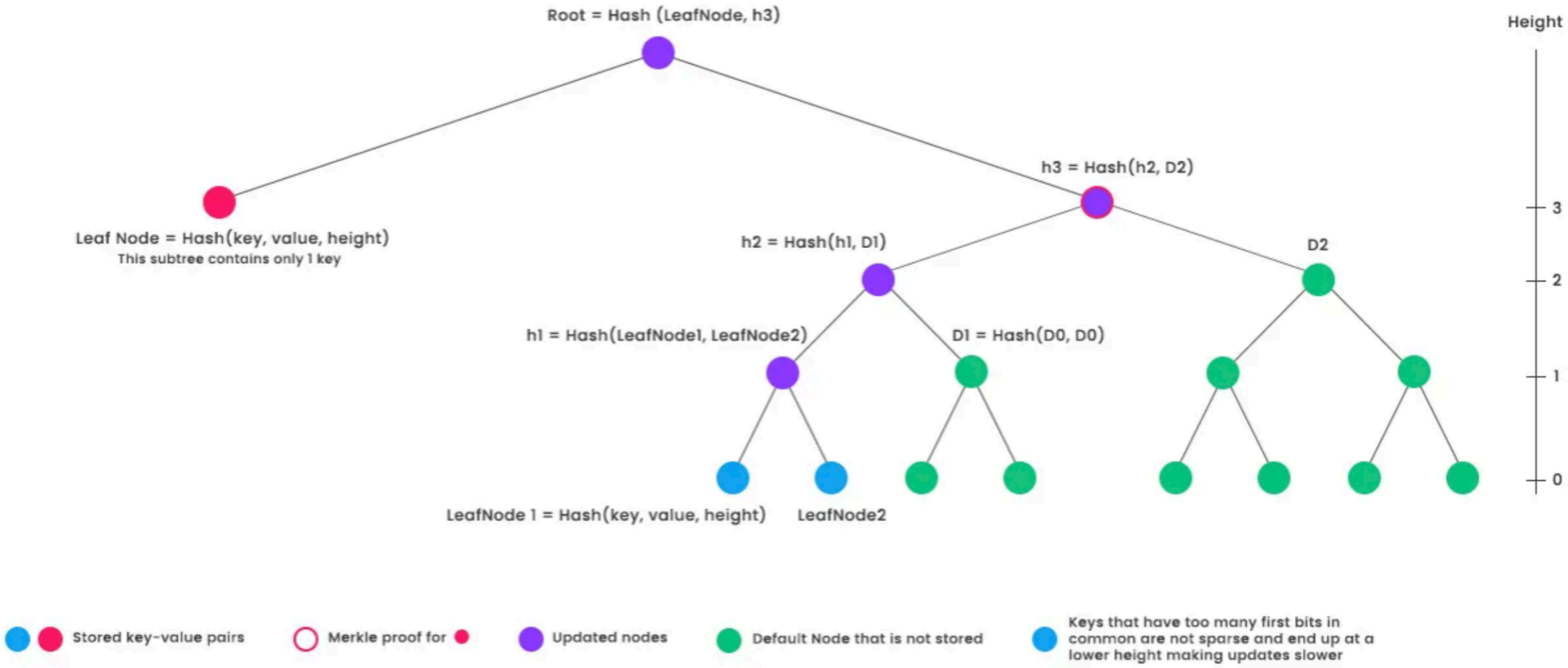
Proof-of-Age Alternatives

Age changes dynamically so two alternatives to manage proof-of-age:

- Dynamic blindable state update when age threshold reached.
- Pre-issued proof-of-age which proof accounts for age change.
  - ZKP predicate proof: ZKP Circuit
  - Sparse Merkle Tree predicate proof:

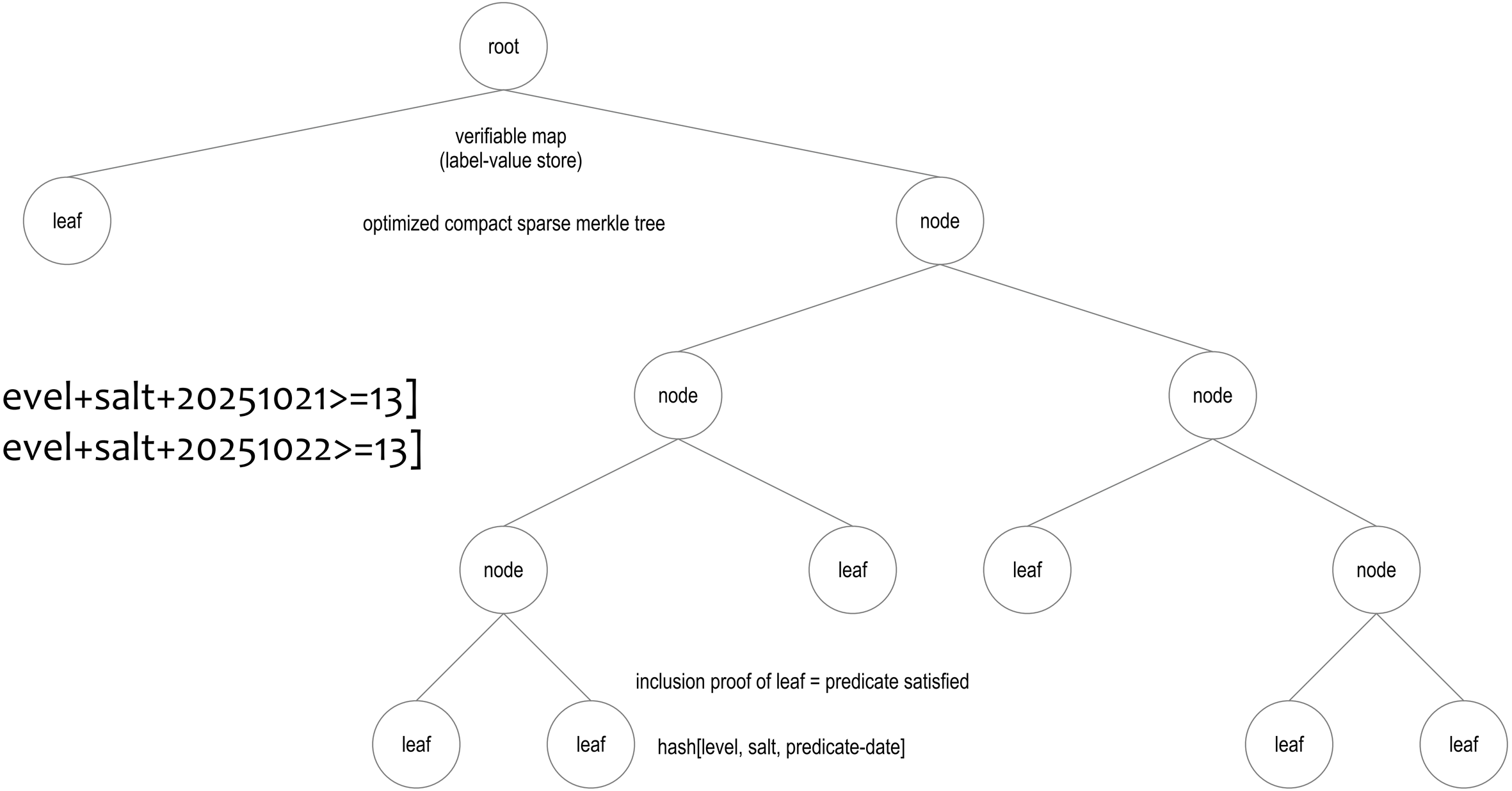
Sparse Merkle Tree

Optimized Compact Sparse Merkle Tree (all operations O(logN))



https://ouvrard-pierre-alain.medium.com/optimized-sparse-merkle-tree-fb481875d56e

Age Predicate Sparse Merkle Tree



H[level+salt+20251021>=13]
H[level+salt+20251022>=13]

IPEX Protocol

Enables graduated disclosure

IPEX Protocol

Discloser	Disclosee	Initiate	Contents	Description
	apply	Y	Schema or its SAID, Attribute field label list, Aggregate element label list, signature on *apply* or its SAID	proposes wanted disclosure
spurn		N		rejects *apply*
offer		Y	Metadata ACDC or its SAID, schema or its SAID, partial disclosure, Aggregate element label list, signature on offer or its SAID	proposes acceptable disclosure
	spurn	N		rejects *offer*
	agree	N	signature and/or anchored seal on offer or its SAID	accepts offer
spurn		N		rejects *agree*
grant		Y	Full or Selective Disclosure ACDC, signature on grant or its SAID	discloses agreed to offer
	admit	N	signature and/or anchored seal on grant or its SAID	confirms received grant disclosure

ZKP to SMT Predicate Proof Comparison

ZKP: relatively small storage, relatively large computation, defeats detectability of issuer key compromise

SMT: relatively large storage, relatively small computation, enables detectability of issuer key compromise

SMT leaf hash is are current datetime value is predicate >= age

Size of SMT: each hash 32 bytes:

Number of leaves: 100 years * 365 days = 36,500

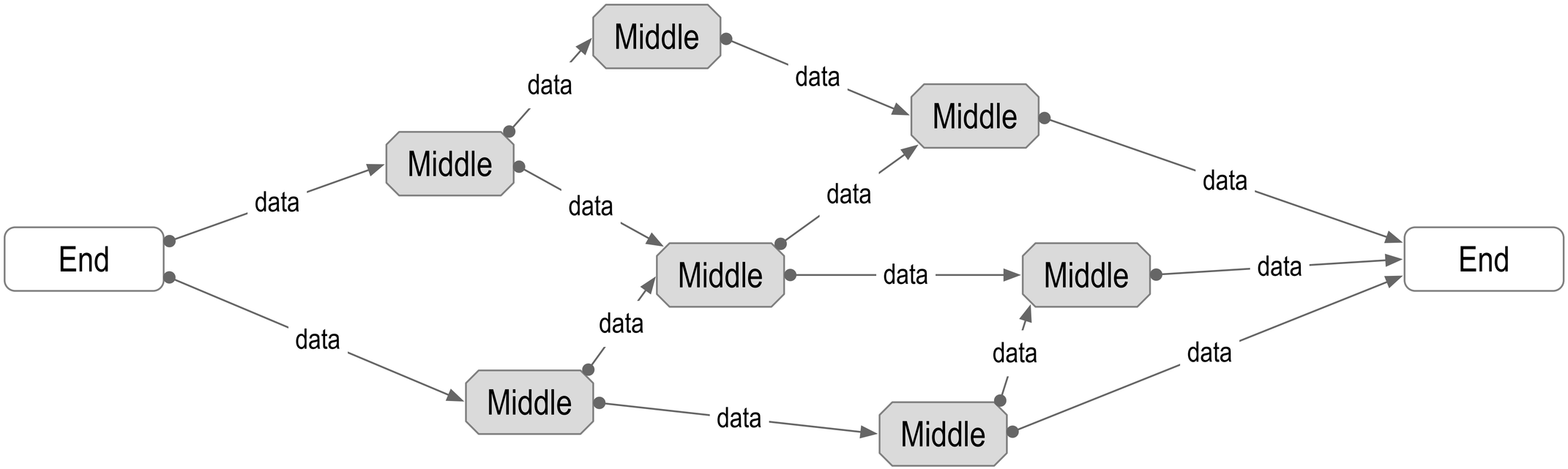
Number of bytes: 32 * 36,500 = 1,168,000

Proof of Inclusion: O(log N) where N is 36,500 = 15 operations

Dual End-Verifiability and End-Only-Viewability

End-to-End Verifiability *of* Authenticity

Only-at-End Viewability via Confidentiality



Ambient Verifiability: any-data, any-where, any-time by any-body

End-only Viewability: one-data, one-where, one-time by one-body

Nonrepudiable *Undeniable*

Each person decides with whom they interact, what they share, and why

Tradespace Threats

Identity Theft

Attention Theft

Intimidation

Tradespace Mitigations

Perpetual Identity

Pepetually Verifiable Issuances

Control over Context

Disclosure, Presence, & Usage

Identity Revolution

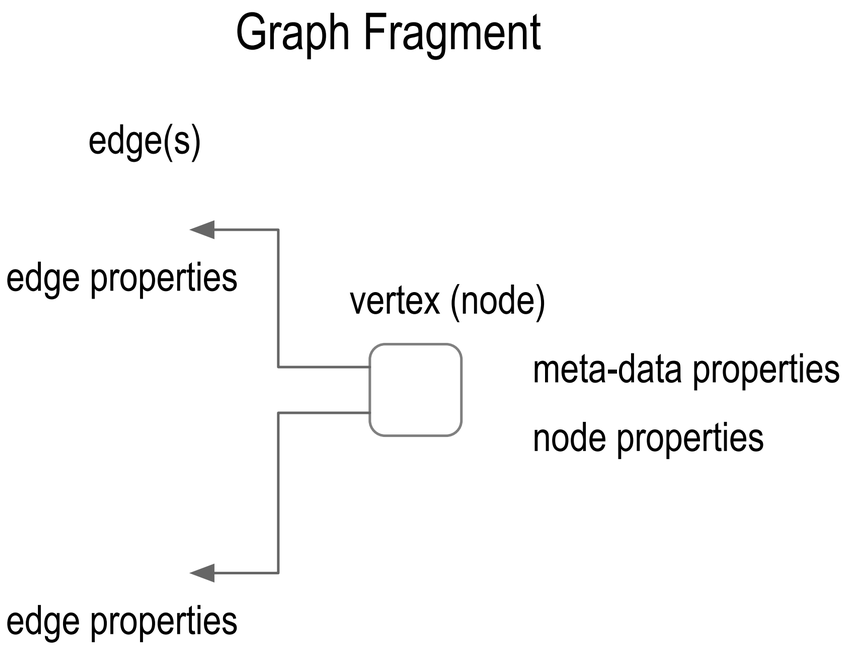
Reputable
Autonomic
Pseudonymity

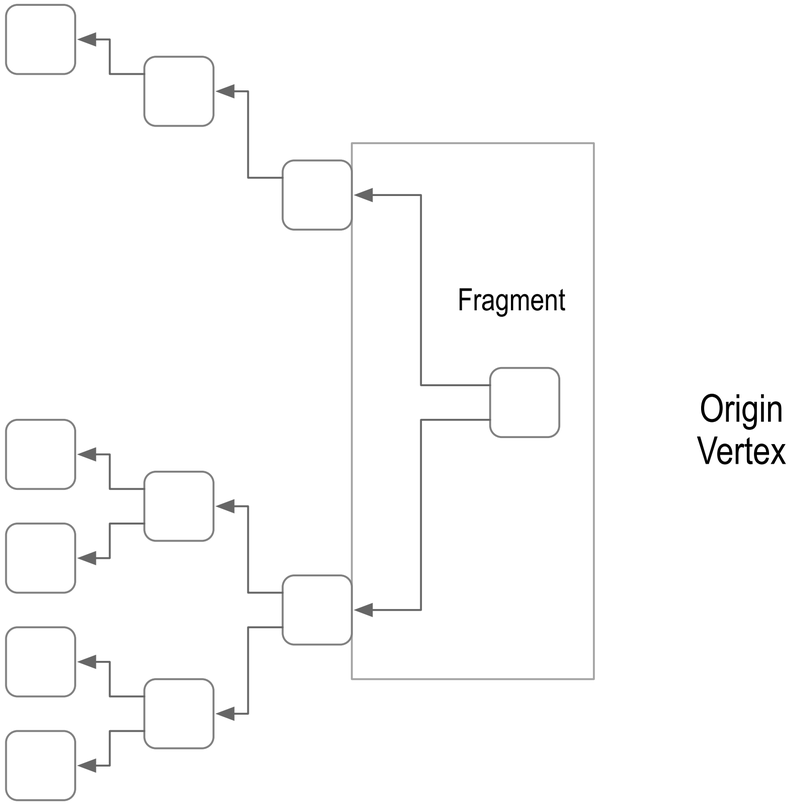
Control
Confidential
Contexts

RAP-C3

Rapsody

Using ACDCs





Perpetually Verifiable Issuances

Each Fragment is Separably Authenticatable

Delegable Attenuable Aggrable Revocable Transferable Authority

High Fidelity Modeling of Real World Authority Structures

Graduated Disclosure (circa 2020)

Compact Disclosure (SAID)

Partial Disclosure

Nested Partial Disclosure

Full Disclosure

Selective Disclosure

Metadata Disclosure (ACDC with some instantiation of nested partial and selective necessary to gain agreement)

Bulk-issued Instance Disclosure (validator-to-validator cryptographic unlinkability)

Combinations of the above

Contractually Protected Disclosure (circa 2020)

Uses graduated disclosure to minimize leakage prior to legally enforceable protection

Chain-Link Confidential Disclosure: strings attached downstream of disclosure, ubiquitous counter-incentive to correlate

Contingent Disclosure:

- Conditions for additional disclosure on disclosee, Escrow disclosure, JIT-NTK disclosure, and Latent Accountability

Legal protection may include retroactive anti-assimilation terms that plug leaks in the graduated disclosure process

Negotiated via IPEX Protocol (issuance and presentation exchange)

Blindable State for Bulk Issuance Protected Disclosure (circa 2023-2025)

Blindable ACDC State Registries

Bindable Blindable Delegated ACDC State Registries

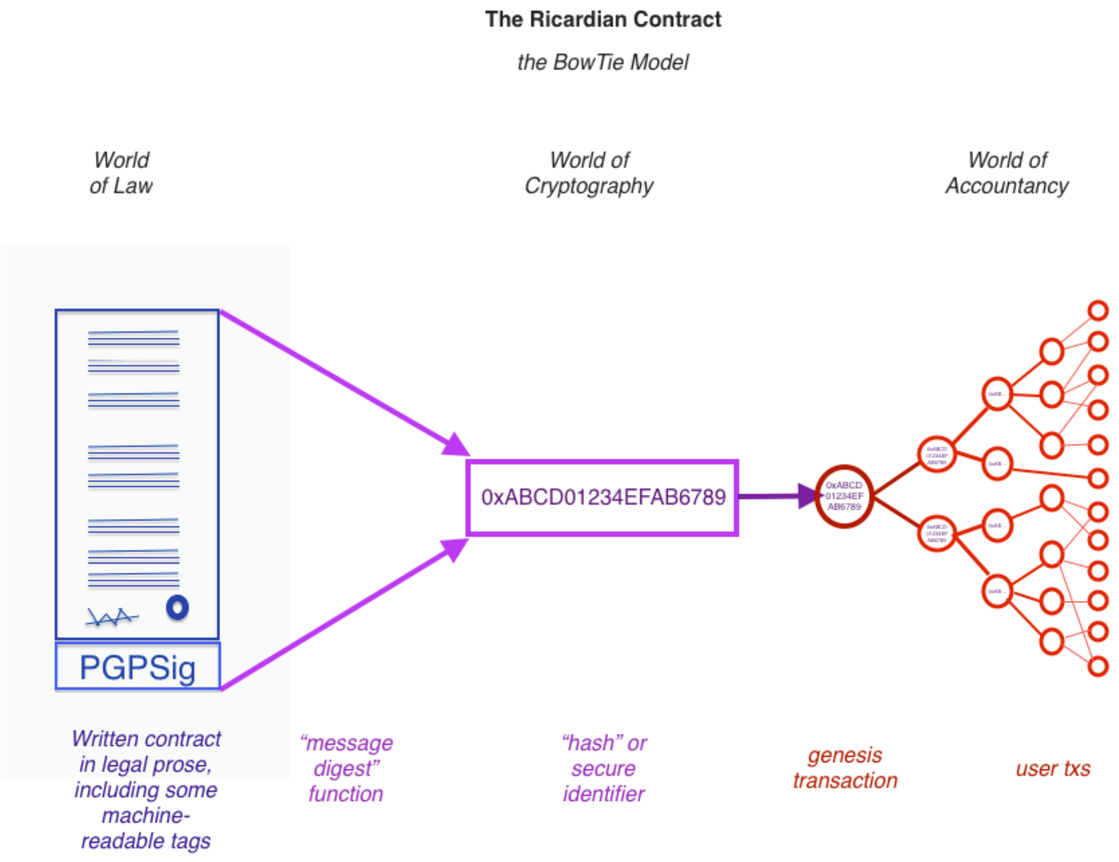
perpetually verifiable, dynamically revocable, transferable, delegable entitlements

Contractually Protected Disclosure

Ricardian Contracts:
https://en.wikipedia.org/wiki/Ricardian_contract

Chain-link Confidentiality
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2045818

Consent, Waiver, Terms-of-use, Remuneration, etc.



Chain-Link Confidentiality

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2045818

A chain-link confidentiality regime contractually links the disclosure of information to obligations to protect that information as the information moves downstream.

The system focuses on the relationships not only between the discloser of information and the initial recipient but also between the initial recipient and subsequent recipients.

Through the use of contracts, this approach links recipients of information as in a chain, with each subsequent recipient bound by the same obligation to protect the information.

These chain contracts contain at least three kinds of terms:

1) obligations and restrictions on the use of the disclosed information;

2) requirements to bind future recipients to the same obligations and restrictions; and

3) requirements to perpetuate the contractual chain.

This approach creates a system for the permissible dissemination of information.

It protects Disclosers by ensuring that the recipient's obligation to safeguard information is extended to third parties.

Chain-Link Confidentiality

Disclosures via Presentations Exchanges may be contractually protected by Chain-Link Confidentiality (i.e a Chain-Link Confidential disclosure).

The chaining in this case is different from the chaining described above between Issuances in a DAG of chained Issuances. Chain-link confidentiality, in contrast, chains together a sequence of Disclosees.

Each Disclosee in the sequence in turn is the Discloser to the next Disclosee.

The terms-of-use of the original disclosure as applied to the original Disclosee MUST be applied by each subsequent Discloser to each subsequent Disclosee via each of the subsequent disclosures (presentation exchanges).

These terms-of-use typically constrain disclosure to only approved parties, i.e. imbue the chain of disclosures with some degree of confidentiality. These terms-of-use are meant to contractually protect the data rights of the original Issuer or Issuee of the data being disclosed.

Contractual Exchange

Discloser provides a non-repudiable ***Offer*** with verifiable metadata (sufficient partial disclosure), which includes any terms or restrictions on use.

Disclosee verifies ***Offer*** against composed schema and metadata adherence to desired data.

Disclosee provides non-repudiable ***Agree*** to terms that are contingent on compliant disclosure.

Discloser provides non-repudiable ***Grant*** disclosure with sufficient compliant detail.

Disclosee responds with ***Admit*** after verifying ***Grant*** using the decomposed schema and disclosed data.

Disclosee may now engage in permissioned use and carries liability as a deterrent against unpermissioned use.

ACDC Normative Field Labels

ACDC Field Labels

Label	Title	Description
		Top-level Fields
v	Version String	Regexable format: `ACDCMmmGggKKKKSSSS.` that provides protocol type, version, CESR genus version, serialization type, size, and terminator
t	Message Type	Three character message type when not default message type
d	Digest (SAID)	Self-referential fully qualified cryptographic digest of enclosing map.
u	UUID	Random Universally Unique IDentifier as fully qualified high entropy pseudo-random string, a salted nonce.
i	Identifier (AID)	Autonomic Identifier whose control authority is established via KERI verifiable Key State.
rd	Registry SAID	Registry for ACDC State.
s	Schema	Either the SAID of a JSON Schema block or the block itself.
a	Attribute	Either the SAID of a block of attributes or the block itself.
A	Aggregate	Either the Aggregate of a selectively disclosable block of attributes or the block itself.
e	Edge	Either the SAID of a block of edges or the block itself.
r	Rule	Either the SAID a block of rules or the block itself.
		Other Reserved Fields
d	Digest (SAID)	Self-referential fully qualified cryptographic digest of enclosing map.
u	UUID	Random Universally Unique IDentifier as fully qualified high entropy pseudo-random string, a salted nonce.
i	Identifier (AID)	Autonomic Identifier whose control authority is established via KERI verifiable Key State.
rd	Registry SAID	Registry for ACDC State.
dt	Datetime	Context dependent ISO datetime string
cargo	Cargo	Field value is embedded encapsulated data
n	Node	SAID of another ACDC as the terminating point of a directed edge that connects the encapsulating ACDC node to the specified ACDC node as a fragment of a distributed property graph (PG).
o	Operator	Either unary operator on edge or m-ary operator on edge-group in edge section. Enables expressing of edge logic on edge subgraph.
w	Weight	Edge weight property that enables default property for directed weighted edges and operators on directed weighted edges.
l	Legal Language	Text of Ricardian contract clause.

Digital information exchange forces hard gradations on “privacy”

Shared or not Shared; Secret or not Secret.

Information that is not-shared = private (one party data)

Information that is shared with restrictions = confidential (two party data, sensitive, protected, secret)

Information that is shared without restrictions = non-confidential (any party data, public)

Surveillance

Non-content meta-data (unrestricted, information about the parties to the interaction)

Content data (confidential)

Content data is confidential, and meta-data about that content is not observable by third parties. (meta-data confidentiality, private network)

Comprehensive Protection Mechanism

The only practically viable protection against the 2nd-party correlation of non-aggregated 1st-party data is pre-disclosure contractual protection by imposing liability on the 2nd party disclosee and strict post-disclosure chain-link confidentiality on any downstream disclosees or other users of that data, including any later assimilation or aggregation.

Chain-link confidentiality requires that 2nd parties not aggregate 1st party data, de-identified or not (selectively disclosed or not), without the consent of the 1st party.

The only sustainable privacy protection mechanism of 1st party data disclosed to a 2nd party, even when selectively disclosed (via ZKP or not), is contractually protected disclosure via chain-link confidentiality.

Any selective disclosure is ineffective unless performed within the confines of a contractually protected disclosure that incentivizes the disclosee (verifier) to protect that disclosure (counter-incentive against the exploitation of that discloser).

Private (19)

adjective

- belonging to some particular person: *private property.*
- pertaining to or affecting a particular person or a small group of persons; individual; personal: *for your private satisfaction.*
- confined to or intended only for the persons immediately concerned; confidential: *a private meeting.*
- personal and not publicly expressed: *one's private feelings.*
- not holding public office or employment: *private citizens.*
- not of an official or public character; unrelated to one's official job or position: *a former senator who has returned to private life;*
  a college president speaking in his private capacity as a legal expert.
- removed from or out of public view or knowledge; secret: *private papers.*
- not open or accessible to the general public: *a private beach.*
- undertaken individually or personally: *private research.*
- without the presence of others; alone: *Let's go into another room where we can be private.*
- solitary; secluded: *He wants to meet us in a more private place.*
- preferring privacy; retiring: *a very private person.*
- intimate; most personal: *private behavior.*
- of, having, or receiving special hospital facilities, privileges, and services, especially a room of one's own and liberal visiting hours: *a private room;*
  a private patient.
- of lowest military rank.
- of, relating to, or coming from nongovernmental sources: *private funding.*

noun

- a soldier of one of the three lowest enlisted ranks.
- **privates.** private parts.

Idioms

- **in private**,
  not publicly; secretly: *The hearing will be conducted in private.*

Confidential (4)

adjective

- spoken, written, acted on, etc., in strict privacy or secrecy; secret: *a confidential remark.*
- indicating confidence or intimacy; imparting private matters: *a confidential tone of voice.*
- having another's trust or confidence; entrusted with secrets or private affairs: *a confidential secretary.*
- (of information, a document, etc.)
  - bearing the classification *confidential,* usually being above *restricted* and below *secret.*
  - limited to persons authorized to use information, documents, etc., so classified.
    Compare classification (def. 5).

Common Law Entick 1765

https://web.archive.org/web/20031021121842/http://www.constitution.org/trials/entick/entick_v_carrington.htm

"Papers are the owner's goods and chattels: they are his ***dearest*** property; and are so far from enduring a seizure, that they will hardly bear an inspection; and though the eye cannot by the laws of England be guilty of a trespass, yet where ***private papers*** are removed and carried away, the ***secret nature*** of those goods will be an aggravation of the trespass, and demand more considerable damages in that respect."

Supreme Court of the US describes Entick thusly:
"a 'great judgment', 'one of the landmarks of English liberty', 'one of the permanent monuments of the British Constitution', and a guide to an understanding of what the Framers meant in writing the Fourth Amendment".

Personal papers cannot be seized; private thought crimes (sedition libel)

US Law

4th and 5th amendments.

Boyd 1886 (unity of 4th and 5th amendments)
“Both amendments relate to the personal security of the citizen. They nearly run into, and mutually throw light upon, each other. When the thing forbidden in the Fifth Amendment, namely, compelling a man to be a witness against himself, is the object of a search and seizure of his private papers, it is an "unreasonable search and seizure" within the Fourth Amendment.”

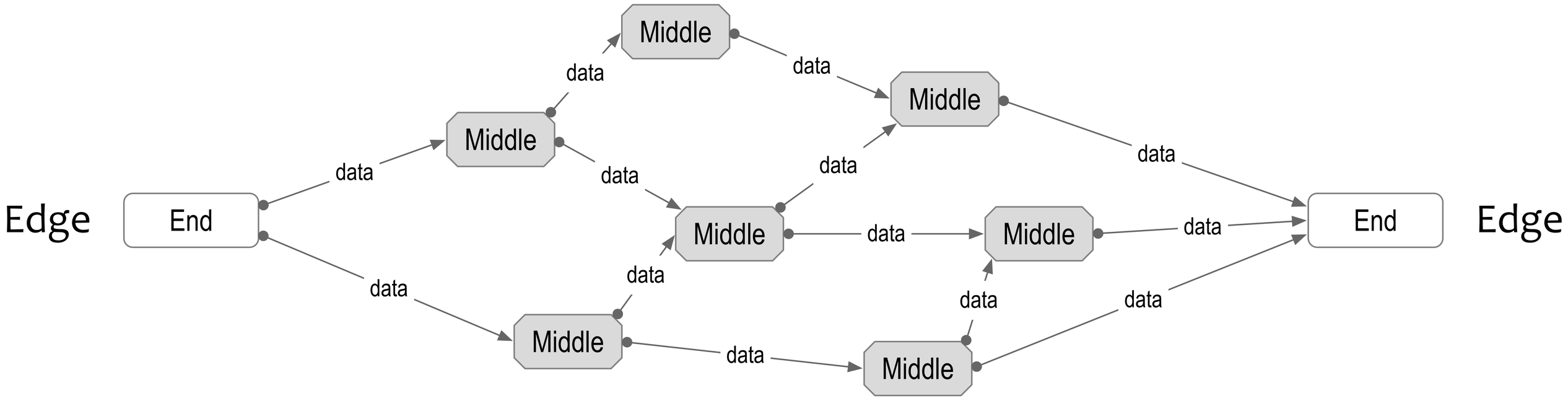
US Law Continued

Katz 1967: Reasonable Expectation of Privacy; Non-content metadata vs content data.

Fisher: 1976: Separation of Fourth and Fifth Amendments. Subpoena power to collect evidence in any recorded form (not direct testimony). Personal private papers are no longer protected.
A court order to reveal secret keys to unlock data creates the "cruel trilemma" by forcing one to decide whether to 1) tell the truth and incriminate one's self, 2) lie and perjure one's self, or 3) refuse to answer and face contempt and jail.

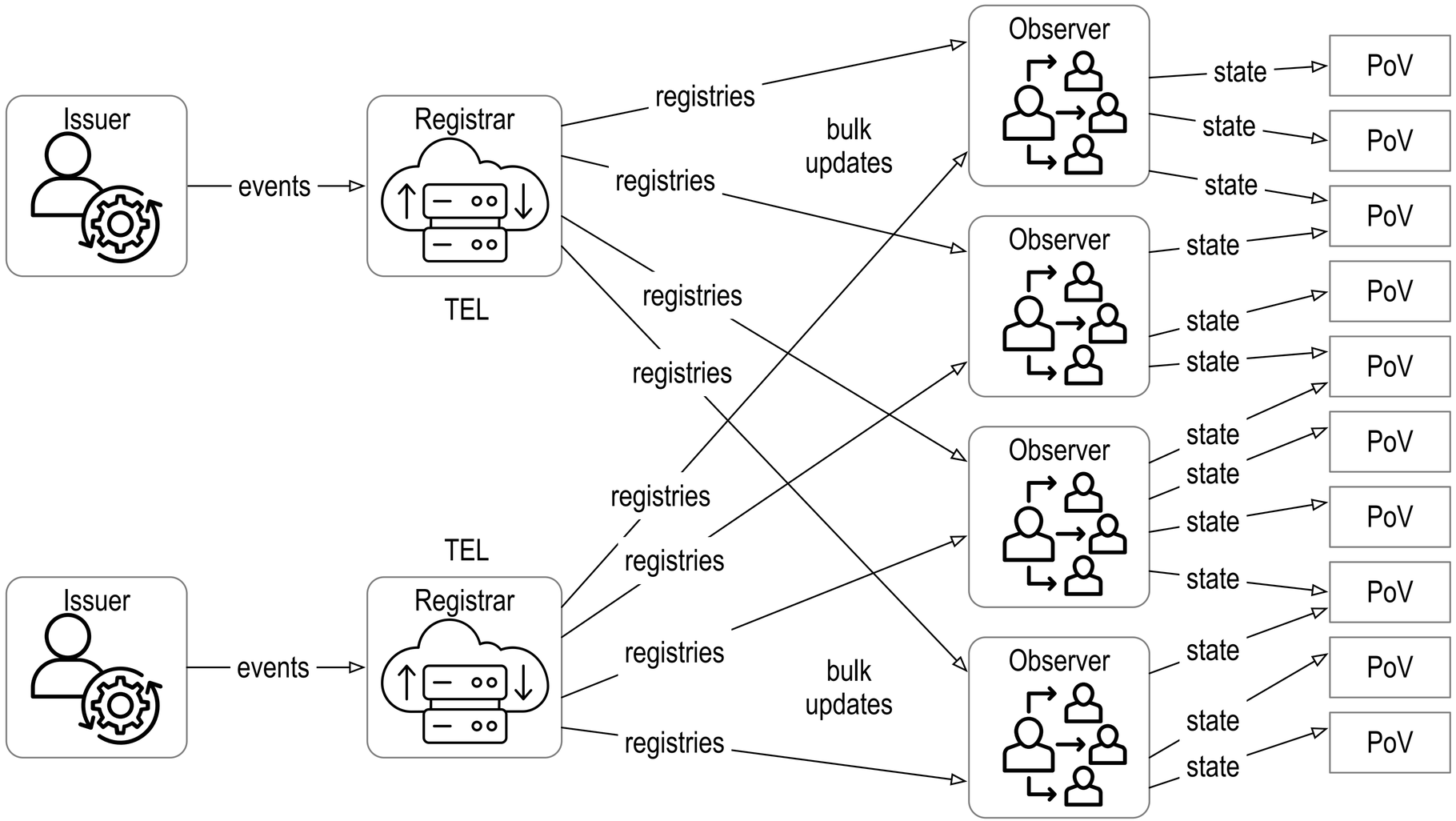
FTC late 1990s: private data rights as consumer rights when promises of notice and consent with respect to shared data are violated (i.e. unfair trade practice).

End-to-End — Edge



With KERI, each AID controller is its own identity provider
Keys-at-the-Edge (KATE) = each edge self-authenticates
No third-party IdPs, no relying parties, no trusted intermediaries

Registrars Observers



ACDC Bound Blinded Registry Example

Registry Inception


```
{
 "v": "ACDCCAACAAJSONAADa.",
  "t": "rip",
  "d": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
  "u": "0ABhY2Rjc3BlY3dvcmtyYXcx",
  "i": "ECWJZFBtllh99fESUOrBvT3EtBujWtDKCmyzDAXWhYmf",
  "n": "0",
  "dt": "2025-07-04T17:51:00.000000+00:00"
}
```


Registry Placeholder Update


```
{
    "v": "ACDCCAACAAJSONAAEi.",
    "t": "bup",
    "d": "EOBVdcIL2rVEzBDpQvmBpsp3R52DsoKhTAAdvHqAz9yc",
    "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
    "n": "1",
    "p": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
    "dt": "2025-08-01T18:06:10.988921+00:00",
    "b": "EFaQ00QW-ZeoMxE9baWcpJbAFXrs5h0ya-wpKnHvMQ0c"
}
```


Virtual Label	Value	Description
d	EFaQ00QW-ZeoMxE9baWcpJbAFXrs5h0ya-wpKnHvMQ0c	BLID (Blinding SAID)
u	aJxtoz6qVeJxPCZvP-qBJifRfIxP3itQBVAAu7JJHxMa	UUID salty nonce blinding factor, HD generated
td	1AAP	Transaction ACDC SAID field value, top-level d
ts	1AAP	Transaction state value string
bn	MAAA	Bound Issuee key event sequence number field
bd	1AAP	Bound Issuee key event SAID field

Unblinded attribute block as attachment to disclosure of TEL Event

-bAaEFaQ00QW-ZeoMxE9baWcpJbAFXrs5h0ya-wpKnHvMQ0caJxtoz6qVeJxPCZvP-qBJifRfIxP3itQBVAAu7JJHxMa1AAP1AAPMAAA1AAP

Placeholder updates provide herd privacy wrt correlation to date time of actual state changes

ACDC Bound Blinded Registry Example …

Registry Issued Update


```
{
    "v": "ACDCCAACAAJSONAAEi.",
    "t": "bup",
    "d": "EGu4B78s6G_GVrzaoBw2a1vkFpB5tVo-wZ1OGsC9D_pK",
    "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
    "n": "2",
    "p": "EOBVdcIL2rVEzBDpQvmBpsp3R52DsoKhTAAdvHqAz9yc",
    "dt": "2020-08-02T12:00:20.000000+00:00",
    "b": "EJAeKLEtVtMtt28IdAKJShyZHodEIZTHJHzaP21A_ZU4"
}
```


Virtual Label	Value	Description
d	EJAeKLEtVtMtt28IdAKJShyZHodEIZTHJHzaP21A_ZU4	BLID (Blinding SAID)
u	aKNPEY4_60x6vUx2g5_5kAoJTn0RDspR04Ql8ecNyTk0	UUID salty nonce blinding factor, HD generated
td	EMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4M	Transaction ACDC SAID field value, top-level d
ts	0Missued	Transaction state value string
bn	MAAB	Bound Issuee key event sequence number field
bd	EJOnAKXGaSyJ_43kit0V806NNeGWS07lfjybB1UcfWsv	Bound Issuee key event SAID field

Unblinded attribute block as attachment to disclosure of TEL Event


```
-bAvEJAeKLEtVtMtt28IdAKJShyZHodEIZTHJHzaP21A_ZU4aKNPEY4_60x6vUx2g5_5kAoJTn0RDspR04Ql8ecNyTkOEMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4M0MissuedMAABEJOnAKXGaSyJ_43kit0V806NNeGW
S07lfjybB1UcfWsv
```

ACDC Bound Blinded Registry Example …

Registry Issued Update


```
{
    "v": "ACDCCAACAAJSONAAEi.",
    "t": "bup",
    "d": "EDXGqM-V_sfj65Dk-lgMnpOZ9DYziqMaYkrju7NRKpFn",
    "rd": "ECOWJI9kAjpCFYJ7RenpJx2w66-GsGlhyKLO-Or3qOIQ",
    "n": "3",
    "p": "EGu4B78s6G_GVrzaoBw2a1vkFpB5tVo-wZ1OGsC9D_pK",
    "dt": "2020-08-03T12:00:20.000000+00:00",
    "b": "EIBNZ3t5rA_-PbBNmhtvtf0VgHBjVrE0fc-DO67f-wGv"
}
```


Virtual Label	Value	Description
d	EIBNZ3t5rA_-PbBNmhtvtf0VgHBjVrE0fc-DO67f-wGv	BLID (Blinding SAID)
u	aFudXE-d0b2owzZNBjd78sx4kCTJx-RTP_Zd19HRUcVD	UUID salty nonce blinding factor, HD generated
td	EMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4M	Transaction ACDC SAID field value, top-level d
ts	Yrevoked	Transaction state value string
bn	MAAI	Bound Issuee key event sequence number field
bd	EDeCPBTHAt75Acgi9PfEciHFnc1r2DKAno3s9_QIYrXk	Bound Issuee key event SAID field

Unblinded attribute block as attachment to disclosure of TEL Event


```
-bAvEIBNZ3t5rA_-PbBNmhtvtf0VgHBjVrE0fc-DO67f-wGvaFudXE-d0b2owzZNBjd78sx4kCTJx-
RTP_Zd19HRUcVDEMLjZLIMlfUOoKox_sDwQaJO-0wdoGW0uNbmI28Wwc4MYrevokedMAAIEDeCPBTHAt75Acgi9PfEciHFnc1r2DKAno3s9_QIYrXk
```